# Solve with us 1.1 - Not Graded

This assignment will not be graded and is only for practice.

**Key Points:**

- Vectors in $R^2$ R2 are represented by ordered pairs $(a, b)$(a, b).
- The first element in the ordered pair denotes the $X$X-coordinate and the second one denotes the $Y$Y -coordinate in the Cartesian plane.
- The addition of two vectors $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2) \in R^2$ v1 = (x1, y1) and v2 = (x2, y2) ) ∈ R2, is given by $v_1 + v_2 = (x_1 + x_2, y_1 + y_2)$v1 + v2 = (x1 + x2, y1 + y2).
- The scalar multiplication of a vector $v = (x, y) \in R^2$ v = (x, y) ∈ R2 with a scalar $c \in R$c ∈ R, is given by $cv = (cx, cy)$cv = (cx, cy).

*1 point*

Let $v_1 = (2, -4)$v1 = (2, −4) and $v_2 = (-1, 2)$v2 = (−1, 2) be two vectors in $R^2$ R2. Which of the following options are true?

[Hint: Recall that vector addition and scalar multiplication are done coordinatewise.]

- ☐ $2v_2 = (-2, 4)$2v2 = (−2, 4)
- ☐ $\frac{1}{2}v_1 = (1, -2)$21v1 = (1, −2)
- ☐ $v_1 + 2v_2 = (0, 0)$v1 + 2v2 = (0, 0)
- ☐ $2v_1 + v_2 = (0, 0)$2v1 + v2 = (0, 0)

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$2v_2 = (-2, 4)$2v2 = (−2, 4)

$\frac{1}{2}v_1 = (1, -2)$21v1 = (1, −2)

$v_1 + 2v_2 = (0, 0)$v1 + 2v2 = (0, 0)

*1 point*

Choose the set of correct options using Figure M2W1SWU1.

[Hint: Recall that vector addition and scalar multiplication are done coordinate-wise.]

- ☐ $2A$2A is the vector (2, 42, 4).
- ☐ $3B$3B is the vector (6, 96, 9).
- ☐ $A + B$A + B is the vector (3, 53, 5).
- ☐ $A - B$A − B is the vector (−1, −1−1, −1).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$2A$2A is the vector (2, 42, 4).

$3B$3B is the vector (6, 96, 9).

$A + B$A + B is the vector (3, 53, 5).

$A - B$A − B is the vector $(-1, -1-1, -1)$.

**2. Key points:**

- The addition of two vectors $V_1 = (a_1, b_1, C_1)$V1 = (a1, b1, C1) and $V_2 = (a_2, b_2, C_2)$V2 = (a2, b2, C2) are done coordinate-wise, as follows:
- $V_1 + V_2 = (a_1, b_1, C_1) + (a_2, b_2, C_2) = (a_1 + a_2, b_1 + b_2, C_1 + C_2)$V1 + V2 = (a1, b1, $C_1$) + (a2, b2, C2) = (a1 + a2, $b_1$ + b2, C1 + C2)

- The scalar multiplication of a vector $V = (a, b, c)$V = (a, b, c) by a scalar $\alpha$α is given by:
$\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$α(a, b, c) = (αa, αb, αc)

*1 point*

Let $V_1 = (1, 0, 0)$V1 = (1, 0, 0), $V_2 = (0, 1, 0)$V2 = (0, 1, 0) and $V_3 = (0, 0, 1)$V3 = (0, 0, 1) be three vectors and $a, b,$ and $c$a, b, and c be three real numbers (scalars). Then, which of the following is (are) true?

☐ $(a, b, c) = aV_1 + bV_2 + cV_3$(a, b, c) = aV1 + bV2 + cV3

☐ $(a, b, c) = abV_1 + bcV_2 + caV_3$(a, b, c) = abV1 + bcV2 + caV3

☐ $(a, 0, c) = aV_1 + cV_2 + 0V_3$(a, 0, c) = aV1 + cV2 + 0V3

☐ $(a, 0, c) = aV_1 + 0V_2 + cV_3$(a, 0, c) = aV1 + 0V2 + cV3

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$(a, b, c) = aV_1 + bV_2 + cV_3$(a, b, c) = aV1 + bV2 + cV3
$(a, 0, c) = aV_1 + 0V_2 + cV_3$(a, 0, c) = aV1 + 0V2 + cV3

**3. Key Points:**

- Addition of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = a11a21a31a12a22a32a13a23a33 and

$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$B = b11b21b31b12b22b32b13b23b33 is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$A + B = a11 + b11a21 + b21a31 + b31a12 + b12

a22 + b22a32 + b32a13 + b13a23 + b23a33 + b33]

- If $R = [abc]$R = [abc] and $C = [\begin{matrix} d \\ e \\ f \end{matrix}]$C = [def], then the product $RC$RC is given by

  $[ad + be + cf]$[ad + be + cf]

- Suppose $A = [\begin{matrix} R_1 \\ R_2 \end{matrix}]$A = [R1R2] and $B = [C_1\ C_2\ C_3]$B = [C1C2C3], where $R_i$ Ri's denote the rows of matrix $A$A and $C_i$ Ci's denote the columns of matrix $B$B. Moreover, assume that the number of columns of $A$A and the number of rows of $B$B are the same. The product $AB$AB is given by,

  $AB = [\begin{matrix} R_1C_1 & R_1C_2 & R_1C_3 \\ R_2C_1 & R_2C_2 & R_2C_3 \end{matrix}]$AB = [R1C1R2C1R1C2R2C2R1C3R2C3]

- Addition of two matrices $A$A and $B$B is defined if both $A$A and $B$B have the same number of rows and the same number of columns. If both the matrices $A$A and $B$B have $m$m rows and $n$n columns, then the matrix $A + B$A + B also has $m$m rows and $n$n columns

- If $A$A is an $m \times n$m × n matrix and $B$B is an $n \times p$n × p matrix, then $AB$AB is well-defined and is an $m \times p$m × p matrix.

*1 point*

Suppose $P = [\begin{matrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{matrix}]$P = [3−17401 2−52], $Q = [14 -9]$Q = [14−9], $R = [0 -3 10]$R = [0−310], $D = [\begin{matrix} -2 \\ 4 \\ 5 \end{matrix}]$D = [−245]

[Hint: If $A$A is a matrix of order $m \times n$m × n and $B$B is a matrix of order $n \times p$n × p, then the order of $AB$AB is $m \times p$m × p.]

- ☐ The matrix $PD$PD is of order $3 \times 1$3 × 1.
- ☐ The matrix $PD$PD is of order $1 \times 3$1 × 3.
- ☐ The matrix $QD$QD is of order $3 \times 3$3 × 3.
- ☐ The matrix $QD$QD is of order $1 \times 1$1 × 1.
- ☐ The matrix $DQ$DQ is of order $3 \times 3$3 × 3.
- ☐ The matrix $DQ$DQ is of order $1 \times 1$1 × 1.
- ☐ The product $QD$QD is not defined.
- ☐ The product $QR$QR is not defined.

The addition $P + Q$P + Q is not defined.

☐

The addition $P + D$P + D is not defined.

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

• P is a 3 × 3 matrix, D is a 3 × 1 matrix. Think about the order of P D.
• Q is a 1 × 3 matrix, D is a 3 × 1 matrix. Think about the order of QD and DQ.
• Q is a 1 × 3 matrix, R is a 1 × 3 matrix. Think whether is it possible to define QR or not.
• Figure out which pair of matrices has the same number of rows as well as the same number of columns.

**Accepted Answers:**

The matrix $PD$PD is of order $3 \times 1$.$3 \times 1$.
The matrix $QD$QD is of order $1 \times 1$.$1 \times 1$.
The matrix $DQ$DQ is of order $3 \times 3$.$3 \times 3$.
The product $QR$QR is not defined.
The addition $P + Q$P + Q is not defined.
The addition $P + D$P + D is not defined.

*1 point*

Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 3 & 4 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 3 \\ 2 & 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 3 \\ 2 & 5 & 4 \end{bmatrix}$B = $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 5 \\ 3 & 3 & 4 \end{bmatrix}$. Which of the following options are correct?

[Hint: Multiplication of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = $\begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{12} & a_{22} & a_{32} & a_{13} & a_{23} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$B = $\begin{bmatrix} b_{11} & b_{21} & b_{31} & b_{12} & b_{22} & b_{32} & b_{13} & b_{23} & b_{33} \end{bmatrix}$ is given by

$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix}$AB = $\begin{bmatrix}$ a11b11 + a12b21 + a13b31a21b11 + a22b21 + a23b31a31b11 + a32b21 + a33b31a11b12 + a12b22 + a13b32a21b12 + a22b22 + a23b32a31b12 + a32b22 + a33b32a11b13 + a12b23 + a13b33a21b13 + a22b23 + a23b33a31b13 + a32b23 + a33b33 $\end{bmatrix}$ Similarly, $BA$BA can be calculated. ]

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$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 26 & 29 \\ 11 & 29 & 34 \end{bmatrix}$AB = $\begin{bmatrix} 5 & 10 & 11 & 10 & 26 & 29 & 11 & 29 & 34 \end{bmatrix}$

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$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 11 & 29 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 25 & 60 \\ 63 & 174 \end{bmatrix}$

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 9 \\ 14 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 64 & 63 \\ 102 & 142 \end{bmatrix}$

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 22 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 9 \\ 14 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 64 & 63 \\ 92 & 141 \end{bmatrix}$

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
Observe that B = $A^T$ (where $A^T$ denotes the transpose of A).
**Accepted Answers:**

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 11 & 29 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 25 & 60 \\ 63 & 174 \end{bmatrix}$

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 9 \\ 14 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 64 & 63 \\ 102 & 142 \end{bmatrix}$

*1 point*
Suppose $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ A = [100−1] and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ B = [0010] Which of the following options are true?

[Hint: Calculate $A^2$ A2 and $B^2$ B2]

☐
$A^2 = I$ A2 = I
☐
$A^2 = A$ A2 = A
☐
$B^2 = I$ B2 = I
☐
$B^2 = 0$ B2 = 0
**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
Correctly solving this question will give a hint to solve Week 4 Lecture 1.2 :
Activity Question 6.
**Accepted Answers:**
$A^2 = I$ A2 = I
$B^2 = 0$ B2 = 0

**4. Key points:**

- Consider a system of linear equations as follows:
$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
a11x1 + a12x2 + a13x3 = b1a21x1 + a22x2 + a23x3
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$
= b2a31x1 + a32x2 + a33x3 = b3

Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11 a21a31a12a22a32a13a23a33],
$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$b = [b1b2b3].

- $A$A is called Coefficient matrixCoefficient matrix.

- Suppose there are $m$m number of equations and $n$n number of variables in the system of linear equations, then the coefficient matrix will be an $m \times n$m × n matrix, $x$x will be an $n \times 1$n × 1 and $b$b will be an $m \times 1$m × 1 matrix.
*1 point*
Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$
−2x1 + 3x2 + x3−x1 + x32x2= 1= 0= 5

If the matrix representation of system (1) is $Ax = b$Ax = b, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], then

[Hint: If a variable is absent in an equation then the coefficient of the variable must be taken as 0.]

☐ $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020]

☐ $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$b = [501]

☐ $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$A = [−2−103021110], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

☐

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020]

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

*1 point*
Consider a system of equations:

$$\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$$ 2x1 + 3x2−2x1 + kx24x1 + 6x2 = 6 = d = 12

Choose the set of correct options.

[Hint: Observe that third equation is a multiple of the first one (Dividing by 2 from both the sides of the third equation gives the first equation). So it is enough to check the solutions for the first and second equation.]

☐ $Ax = b$Ax = b represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$x = [x1x2], $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$A = [2−243k66], and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$b = [6d12]

☐ The system has no solution if $k = -3$k = −3, $d = 0$d = 0.

☐ The system has a unique solution if $k = 3$k = 3, $d = 0$d = 0.

☐ The system has infinitely many solutions if $k = -3$k = −3, $d = 6$d = 6.

☐ The system has infinitely many solutions if $k = -3$k = −3, $d = -6$d = −6.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$Ax = b$Ax = b represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$x = [x1x2], $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$A = [2−243k6], and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$b = [6d12]

The system has no solution if $k = -3$k = −3, $d = 0$d = 0.
The system has a unique solution if $k = 3$k = 3, $d = 0$d = 0.
The system has infinitely many solutions if $k = -3$k = −3, $d = -6$d = −6.

**5. Key points:**

- Let $A$A be a $2 \times 2$2 × 2 matrix as follows:

  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$[acbd]

  $det(A) = ad - bc$det(A) = ad − bc

- Let $A$A be a $3 \times 3$3 × 3 matrix as follows:

  $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$[a11a21a31a12a22a32a13a23a33]

  $det(A) = a_{11} \times det(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}) - a_{12} \times det(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}) + a_{13} \times det(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix})$det(A) = a11 × det ([a22a32a23a33]) − a12 × det ([a21a31a23a33]) + a13 × det ([a21a31a22a32])

  i.e., $det(A) = a_{11} \times (a_{22}\, a_{33} - a_{32}\, a_{23}) - a_{12} \times (a_{21}\, a_{33} - a_{23}\, a_{31}) + a_{13} \times (a_{21}\, a_{32} - a_{22}\, a_{31})$ det(A) = a11 × (a22a33 − a32a23) − a12 × (a21a33 − a23a31) + a13 × (a21a32 − a22a31)
  (Expanding with respect to first row)

Find the determinant of the matrix $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$[3121]

[Hint: Use the formula for finding the determinant of a matrix of order $2 \times 2$2 × 2.]

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*
*1 point*

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$A = [322232211] be a $3 \times 3$3 × 3 matrix. Which of the following is(are) correct? [Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

☐ $det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&2\\1&2\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

☐ $det(A) = -2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) + 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

☐ $det(A) = 2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) - 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = 2 × det ([2121]) − 3 × det ([3221

]) + 2 × det ([3221])

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}2&2\\2&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

$det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&2\\1&2\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

$det(A) = -2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) + 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

**6. Key points:**

- Row operations and relation with determinants:

  - Type 1: Interchanging two rows of a matrix changes the sign of the determinant.
  - Type 2: Multiplying a real number with a row and adding it to some other row does not change the determinant.
  - Type 3: The determinant of a new square matrix obtained by multiplying a real number $c$c with a row of square matrix of order $n$n, is $c$c times the determinant of the original matrix.

- Same types of column operations, as discussed above for row operations, gives same type of relation with the determinants.
- If two rows or two columns of a matrix are equal, then the determinant of the matrix is 00.
- If all the elements of any row or any column of a matrix are 00, then the determinant of the matrix is 0 0.
- $det(A) = det(A^T)$det(A) = det(AT ), where $A^T$ AT denotes the transpose of $A$A.
- $det(AB) = det(BA) = det(A)det(B)$det(AB) = det(BA) = det(A)det(B).

*1 point*
Let $A$A be a $3 \times 3$3 × 3 matrix with non-zero determinant. If $det(2A) = k\ det(A)$det(2A) = k det(A), then what will be the value of $k$k?
[Hint: If a scalar $(c)$(c) is multiplied with one row of a matrix $A$A, then the determinant of the new matrix will be $c$c times the determinant of $A$A.]

○
22
○
44
○
88
○
1616

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
88

*1 point*

Let $A$A be a $2 \times 22 \times 2$ matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$[a11a21a12a22]. Define the following matrices :

$B = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$B = [a11 − a21a21a12 − a22a22], $C = \begin{bmatrix} a_{11} & -a_{12} \\ a_{21} & -a_{22} \end{bmatrix}$C = [a11 − a12a21 − a22a12a22],

$D = \begin{bmatrix} a_{11} & +a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$D = [a11 + a21a21a12 − a22a22], $E = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & +a_{22} \end{bmatrix}$E = [a11 − a21a21a12 + a22a22]

Which of the matrices among $B, C, D,$ and $E$B, C, D, and E have the same determinant as that of the matrix $A$A, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$ a11, a12, a21, a22?
[Hint: Try to observe whether $B, C, D,$ B, C, D, and $E$E can be obtained from $A$A by row or column operations of Type 22 mentioned in the key points.]

○
$B$B and $D$D
○
$B$B and $E$E
○
$B$B and $C$C
○
$D$D and $E$E
○
$C$C and $E$E

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
• B can be obtained by a row operation of Type 2.
• C can be obtained by column operation of Type 2.
• D cannot be obtained by a row operation or column operation of Type 2. Check the determinant of D and see whether it matches with determinant of A or not.
• E cannot be obtained by a row operation or column operation of Type 2. Check the determinant of E and see whether it matches with determinant of A or not.
**Accepted Answers:**
$B$B and $C$C

*1 point*

If all the elements of a $3 \times 33 \times 3$ real matrix $A$A are the same, then which of the following is (are) correct?

[Hint:
∵ If two rows or two columns of a matrix are equal, then the determinant of the matrix is 0.
∵ $det(A) = det(A^T)$det(A) = det(AT), where $A^T$ AT denotes the transpose of $A$A].

- [ ] Determinant of matrix $A$A is 00.
- [ ] Determinant of matrix $A$A cannot be determined from the given information.
- [ ] Determinant of matrix $A$A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$ A + AT is 00, where $A^T$ AT denotes the transpose of $A$A.
- [ ] Determinant of matrix $A + A^T$ A + AT cannot be determined from the given information, where $A^T$ AT denotes the transpose of $A$A.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Determinant of matrix $A$A is 00.
Determinant of matrix $A + A^T$ A + AT is 00, where $A^T$ AT denotes the transpose of $A$A.

*1 point*

Let $A, B,$ A, B, and $C$C be $3 \times 3$3 × 3 real matrices. Which of the following options is (are) correct?
[Hint: ∵ $det(A) = det(A^T)$det(A) = det(AT), where $A^T$ AT denotes the transpose of $A$A.
∵ $det(AB) = det(BA)$det(AB) = det(BA). ]

- [ ] $det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)
- [ ] $det(A^3) = det(A)^3$ det(A3) = det(A)3
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$det(A + B + C) = det(A) + det(B) + det(C)